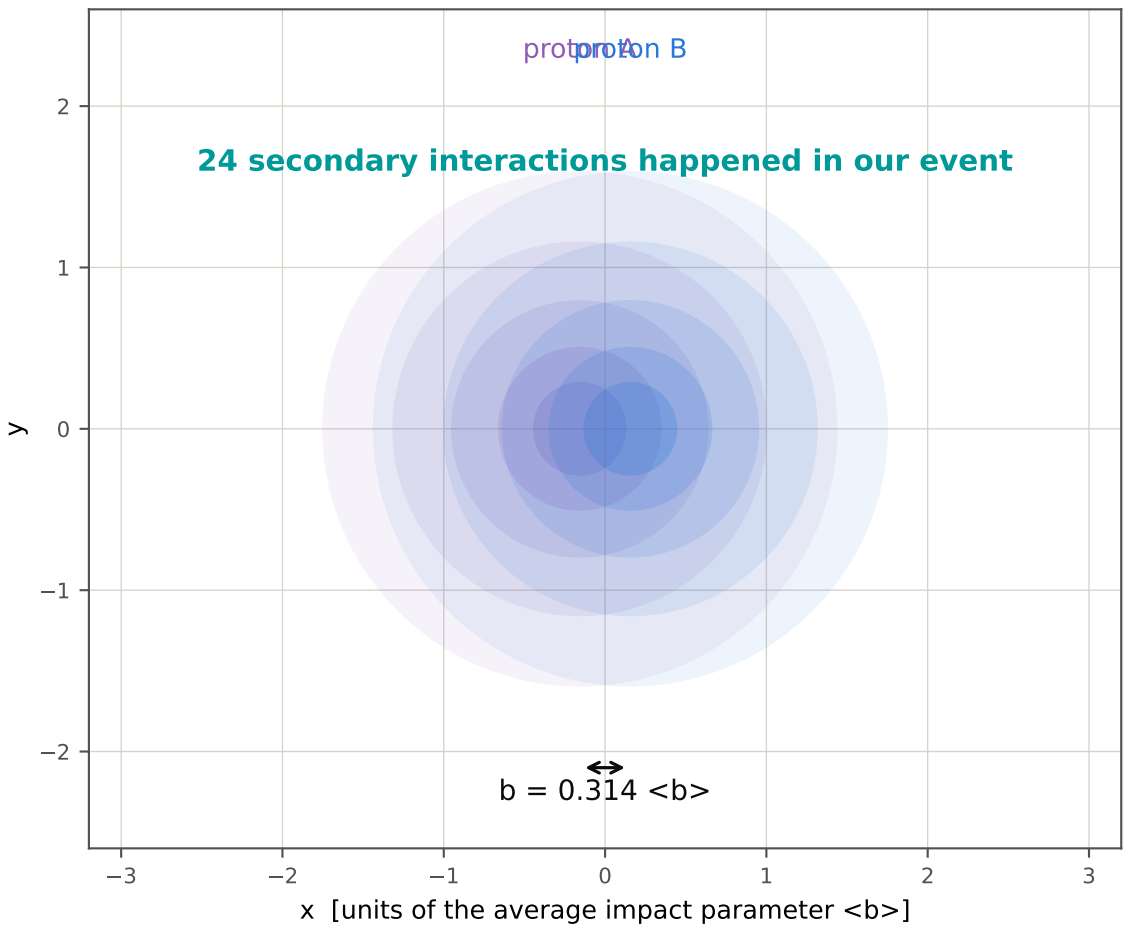
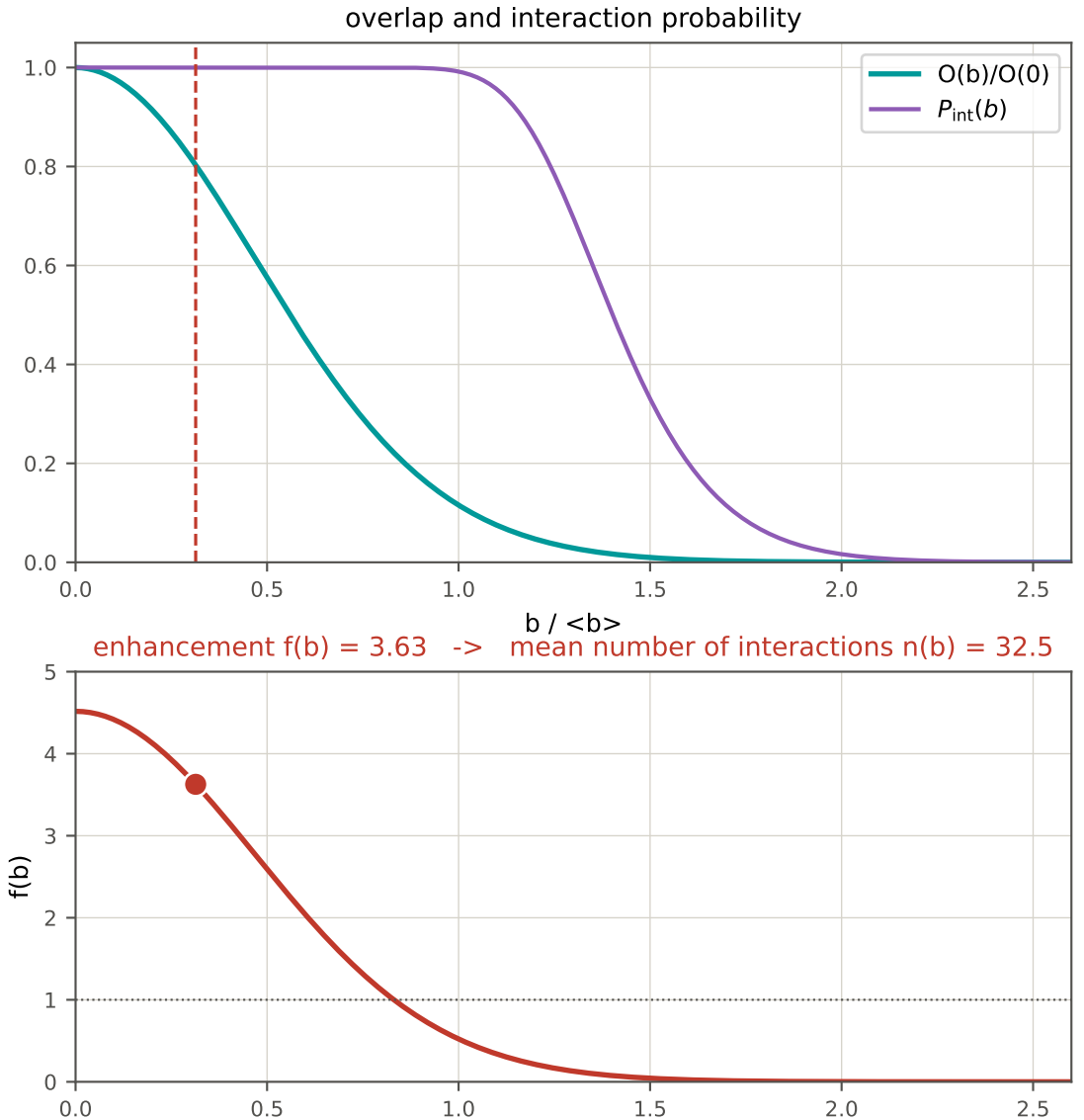
Scene 1: two protons meet at impact parameter b

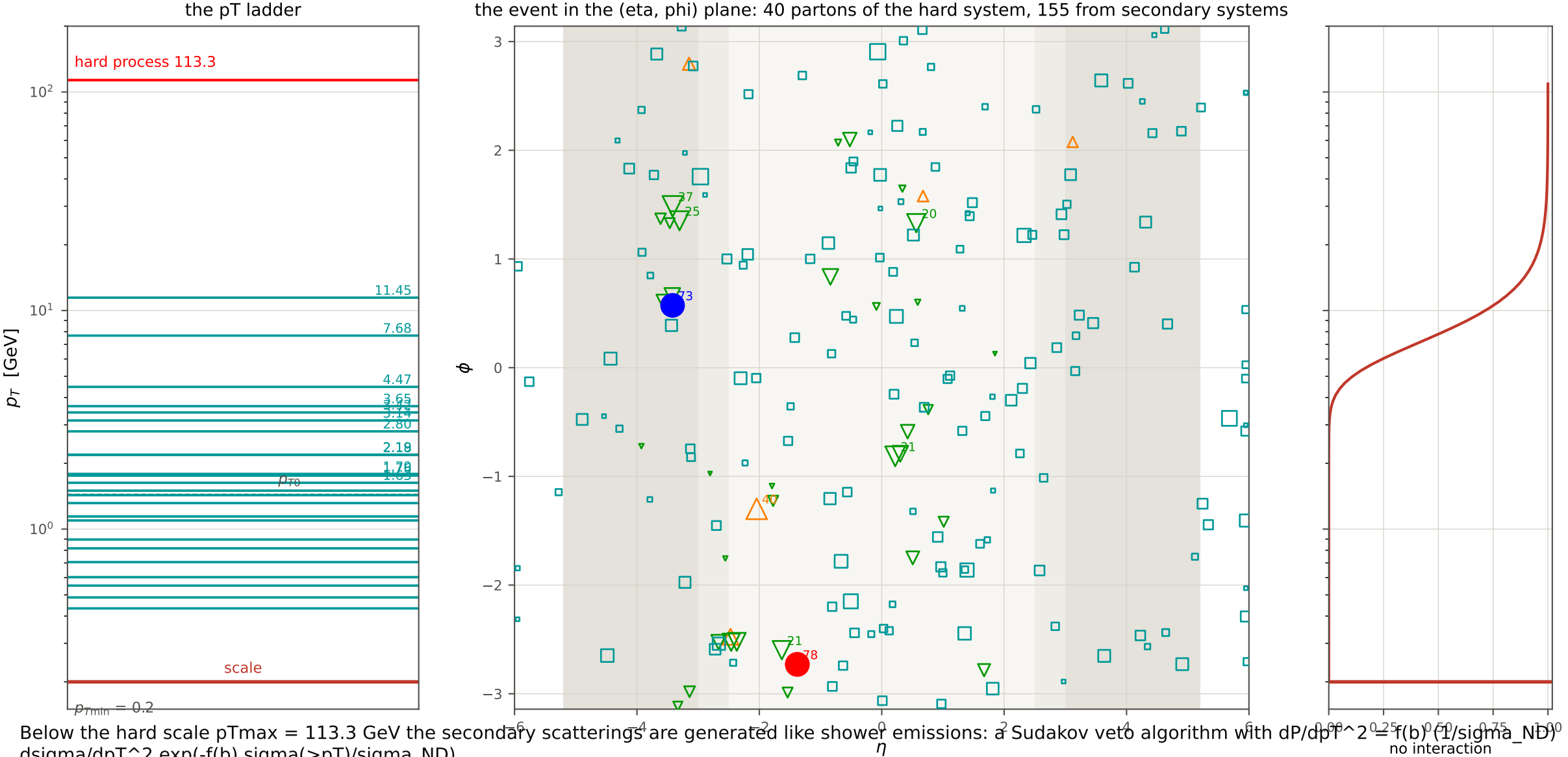
standalone_multiparton_interactions.py





The matter of each proton is a double Gaussian (CP5: 63 % of it in a core of 0.76 of the radius). The chance of a parton-parton interaction is the overlap O(b) of the two profiles; the mean number of interactions is n(b) = f(b) <n> with <n> = sigma_int/sigma_ND = 8.97.
An event with a hard process is more central than average: b is drawn with weight O(b). Our event sits at b = 0.3136 <b>, f = 3.63 (the CMSSW event: 0.3136, 3.709).

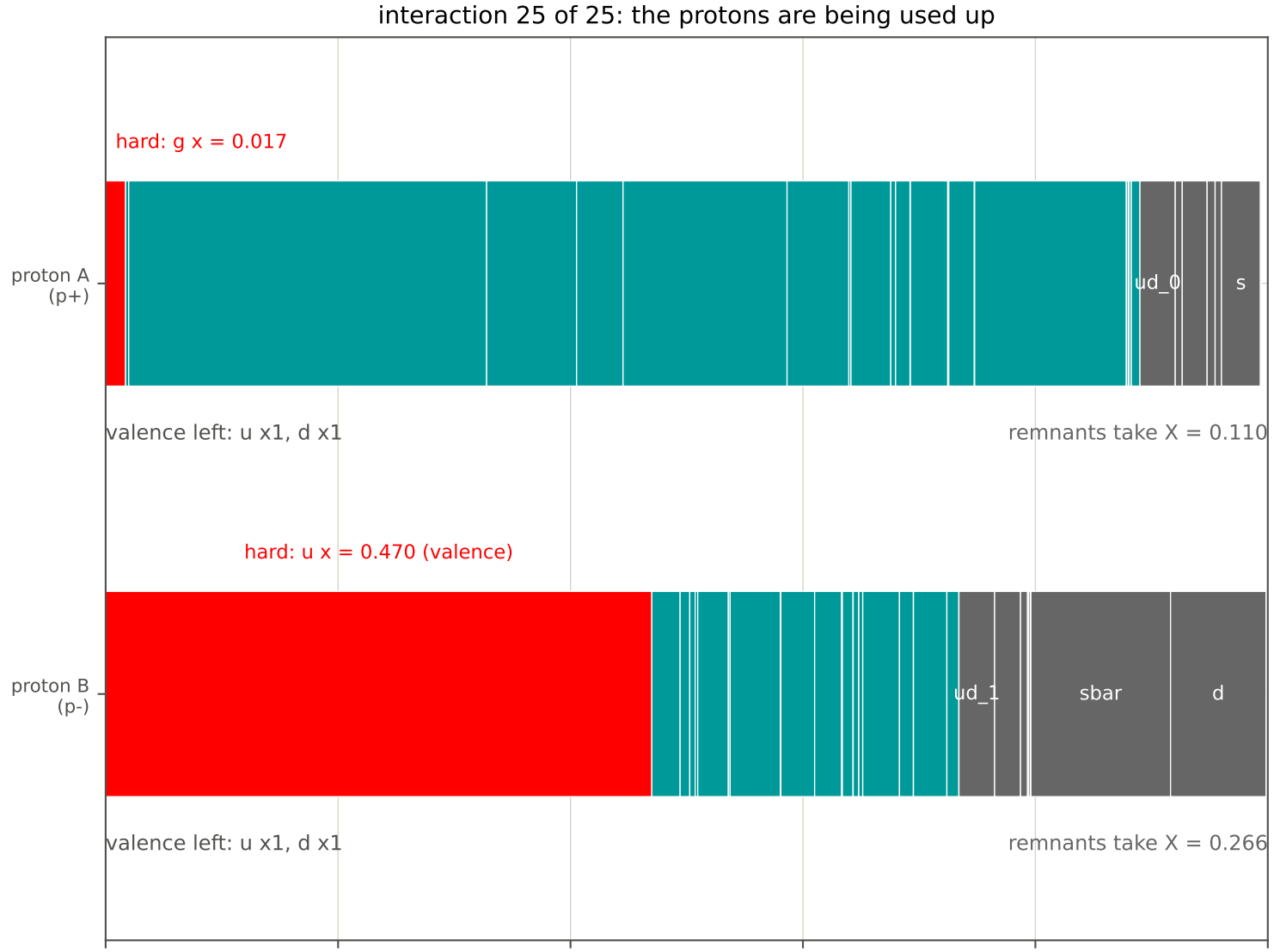
Scene 2: the chain of interactions runs down in pT

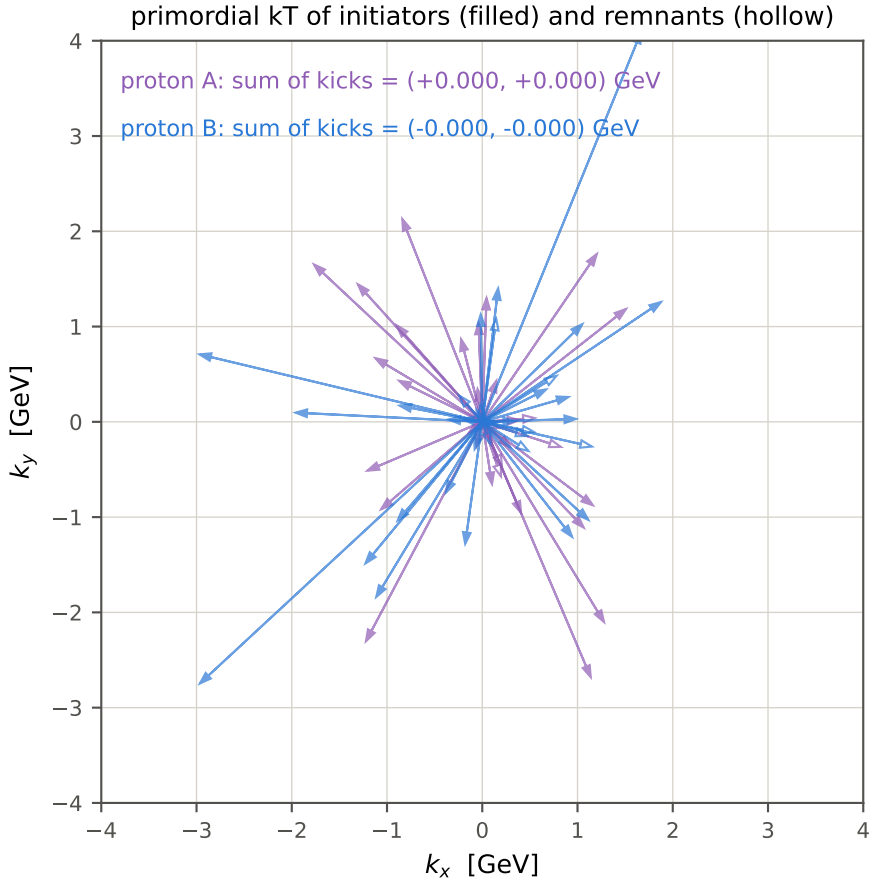


Below the hard scale pTmax = 113.3 GeV the secondary scatterings are generated like shower emissions: a Sudakov veto algorithm with dP/dpT^2 = f(b) (1/sigma_ND) dsigma/dpT^2 exp(-f(b) sigma(>pT)/sigma_ND).
Each accepted scale gives two rapidities, two momentum fractions, two flavours and a process, drawn with the depleted PDFs of the two protons; the new system is then showered from its own pT. Scale now: 0.20 GeV, 24 of 24 interactions happened.

Scene 3: what is left of the protons: the beam remnants

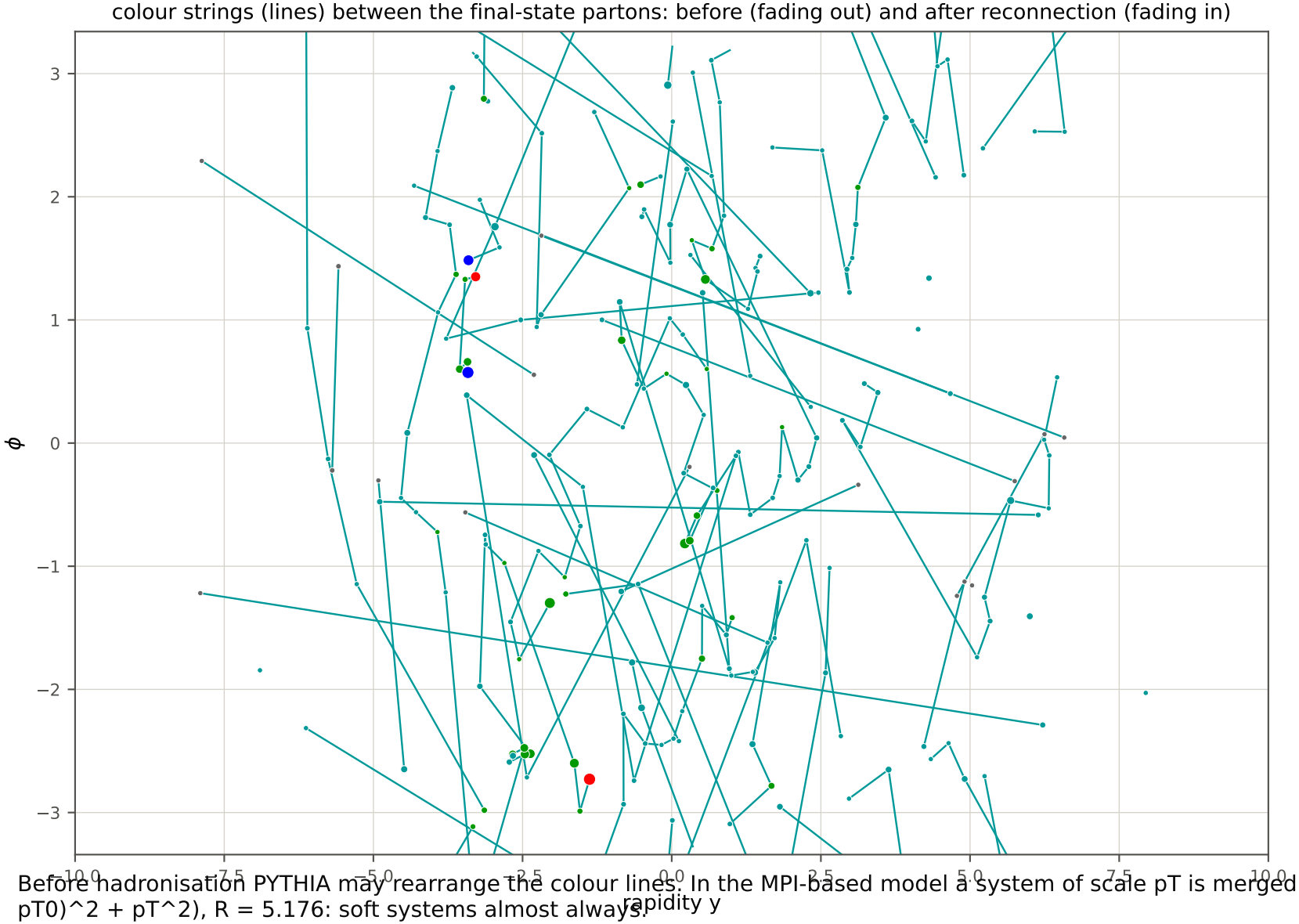
standalone_multiparton_interactions.py

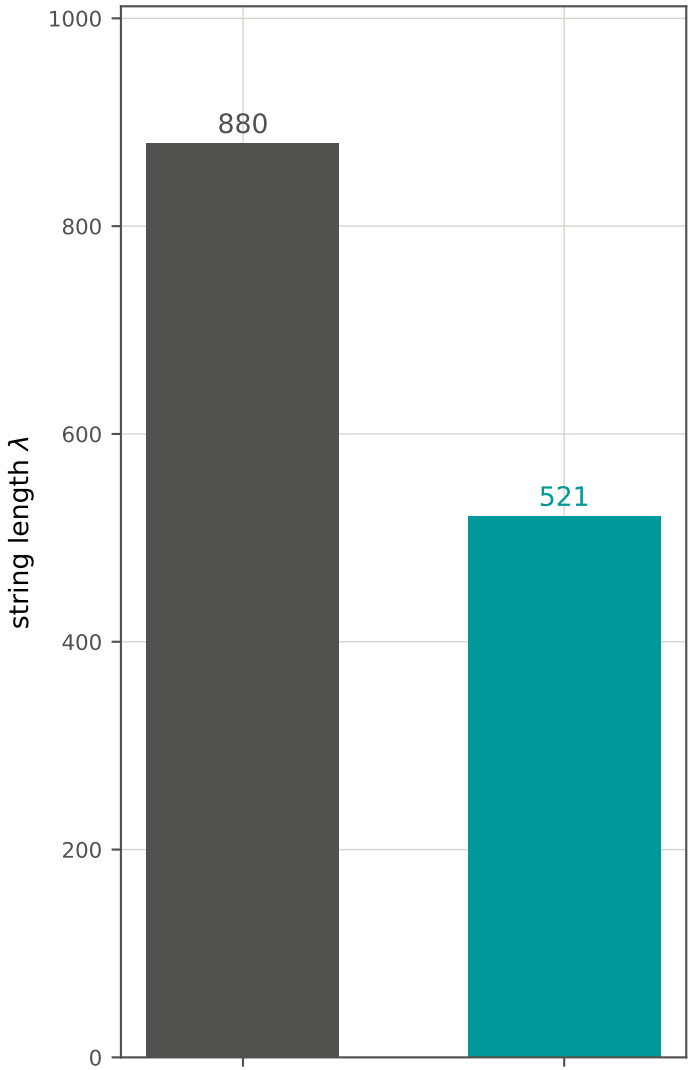




Every interaction takes a momentum fraction x out of a proton; the PDFs are rescaled to what is left, a kicked-out valence quark is subtracted from the valence content and a kicked-out sea quark leaves a companion. Proton A gave away 25 partons (X = 0.110 left), proton B 25 (X = 0.266).
The rest becomes the remnant partons (A: ud_0, u, u, ubar, sbar, s, ubar; B: ud_1, u, u, sbar, u, sbar, ubar, sbar, d); every parton gets a primordial kT and the kicks cancel within each proton.

Scene 4: colour reconnection shortens the strings

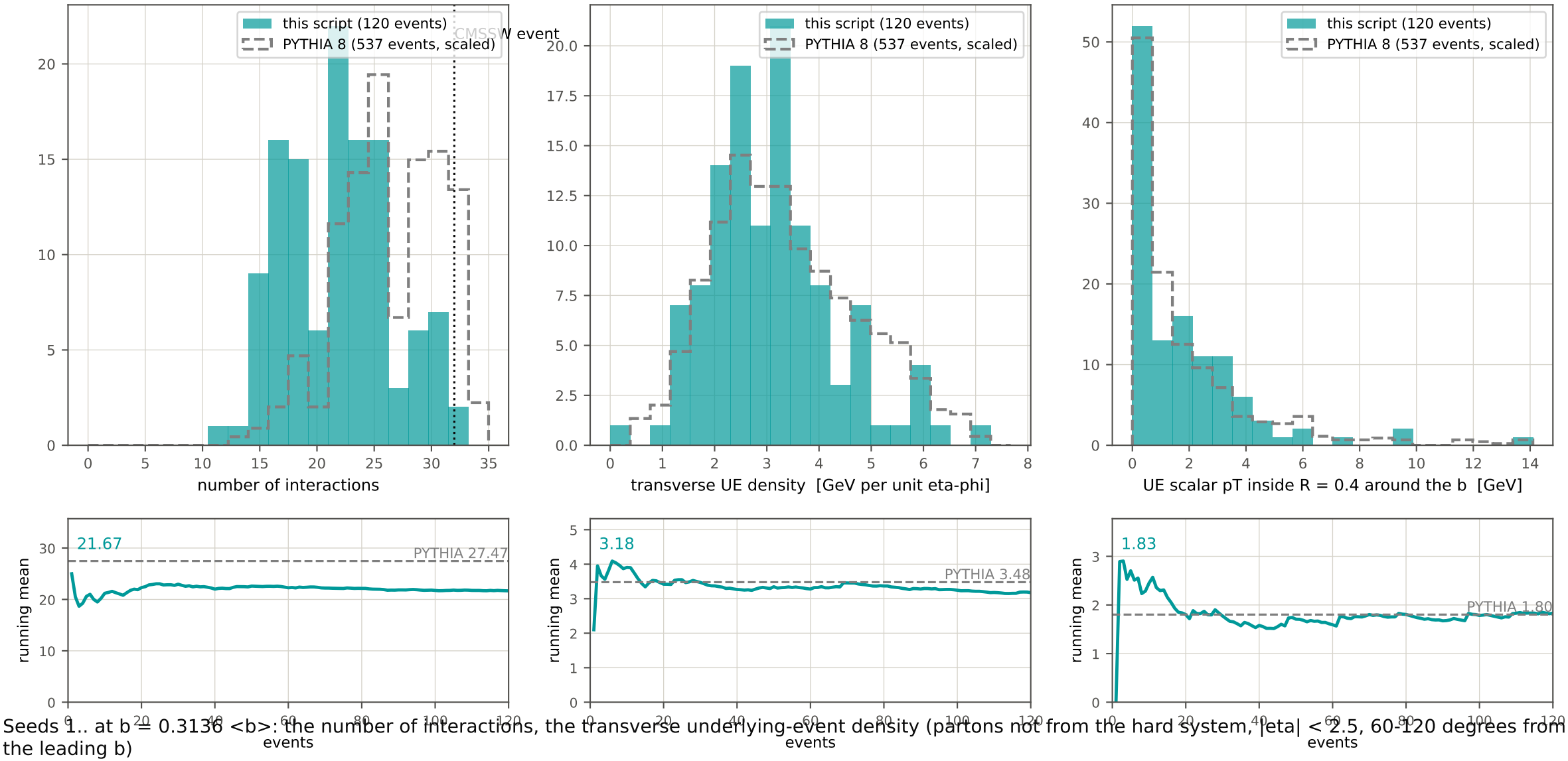




Before hadronisation PYTHIA may rearrange the colour lines. In the MPI-based model a system of scale pT is merged into a harder one with probability P = (R pT0)^2/((R pT0)^2 + pT^2), R = 5.176: soft systems almost always.
Its gluons are inserted into the dipoles of the harder system where they disturb it least, so the total string length lambda = sum ln(1 + m_ij^2/m0^2) drops: 880 -> 521 here, 23 of 24 systems merged; the strings get shorter, not fewer: still 33 colour-singlet strings (one per quark end; the merged gluons only move between them).

Scene 5: the same event dressed again and again

standalone_multiparton_interactions.py



Seeds 1.. at b = 0.3136 <b>: the number of interactions, the transverse underlying-event density (partons not from the hard system, |eta| < 2.5, 60-120 degrees from the leading b)
and the extra scalar pT inside R = 0.4 around the b quark fluctuate strongly from event to event; their means settle towards the values PYTHIA 8 gives for the same hard event (standalone_mpi_stats_fixedb.stats.json).